

Microsoft open sources GW-BASIC

Microsoft has open sourced GW-BASIC on GitHub, 37 years after the first version of it was released for the IBM PC on February 10, 1983. GW-BASIC is a BASIC interpreter, written in Assembly language.



Rich Turner, senior program manager of Windows Console and Command-Line at Microsoft, said in a post, “These sources, as clearly stated in the repo’s readme, are the 8088 assembly language sources from February 10, 1983, and are being open-sourced for historical reference and educational purposes. This means we will not be accepting PRs that modify the source in any way.”

Turner also said that Microsoft’s BASIC implementations can be traced back to Bill Gates and Paul Allen’s implementation of the first product for Microsoft, which was a BASIC interpreter for the Altair 8800. As the source code of GW-BASIC was written fully in Assembly language, there is no source code for it in C.

Assembly language was used back then, instead of high-level languages for software, as the compilers for such languages were often very expensive. They seldom generated efficient code, and there were space and performance constraints with respect to home and personal computers of that time.

Seven in ten applications have a security flaw in an open source library: Veracode

Seven in ten applications have a security flaw in an open source library on initial scan, as per new research by Veracode. This highlighted how the use of open source can introduce flaws, increase risk and add to security debt.



SpaceChain Foundation invests in Core Semiconductor

SpaceChain Foundation has announced that it has contracted and invested in Core Semiconductor to produce the world’s first open source hardware platform, which will be able to provide a downlink to mobile phones and small devices directly from satellites in orbit. This will be done without the use of a satellite dish on Earth or a third party network. It said that with the security inherently built-in, the technology is designed with the blockchain industry in mind and to bring blockchain applications to a global user base.



Jeff Garzik, SpaceChain Foundation co-founder and CTO said, “GPS is a low data rate protocol that has transformed the way we navigate the world and created over US\$ 400 billion in equity value. We want to do the same for emerging digital economies. Adding secure direct downlink and location capabilities to devices on a robust multi-layer global decentralised infrastructure will bring blockchain to mass adoption.”

Core Semiconductor sources have said that the company has designed the platform to be small enough to fit inside any handheld device. They have also said that the platform is affordable and is easy to deploy. The technology has been designed for low bit rate applications of around 1250 bytes per minute, which makes it suitable for verifying blockchain hashes and encrypted signatures. Anyone can verify the security and correctness of the design themselves, directly on GitHub, as the hardware is open source.